Blood Rain: The Sky Mystery With a Surprising Answer

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For thousands of years, people have reported rain that falls red 'blood rain.' Stories of it appear in texts as old as the Iliad and the Bible. Some people saw omens in it. Scientists looked for a natural cause.

In August 1841, something strange fell from the sky near Lebanon, Tennessee. Two people working in a tobacco field heard a pattering on the leaves, followed by what looked like blood raining down. The farm's owner sent samples to the scientist Gerard Troost. He found reddish-brown muscle fibers that he said looked much like dried beef, and he believed the material could only have come from the shredded carcass of an animal. He guessed that a powerful wind had swept up part of a decomposing animal and carried it into a storm cloud. Despite the limits of mid-19th-century science, Troost was partly right: strong winds can lift surprising things.

Today, meteorologists have a more down-to-earth answer for objects falling from the sky. A **waterspout** forms when a column of low-pressure air spins violently over a body of water. Powerful **updrafts** inside its funnel can pull up water, and whatever is in it, then carry it aloft and drop it somewhere else. That could explain reports of frogs and fish raining from the sky, too. In 1841, the closest large body of water to Lebanon was the Cumberland River, several miles to the north, but that distance should not rule out a **waterspout**, because these storms can carry what they pick up for miles. No one can say for certain what happened that day it happened more than a century before weather satellites but a strong **updraft** is a plausible explanation.

But many blood rains have fallen with no flesh at all. That raises a different question: what turns rain red? During the 2001 **monsoon** season, crimson rain fell on and off across Kerala, India. Physicist Godfrey Louis examined samples under a microscope and found what looked like biological cells. He argued that they were alien microbes released by a comet that broke apart in the atmosphere. However, a government-commissioned investigation found that they were **spores** of **Trentepohlia**, a reddish-orange **alga** that grows abundantly on trees, rocks, and even lamp posts in the region. When researchers grew samples of local lichen in the lab, they produced the same **alga** found in the rainwater. The **spores** did not come from space or from thousands of miles away. They most likely came from nearby, released in huge numbers and swept into rain clouds by the wind.

Other blood rains have been traced to windblown red dust, often from the Sahara, which mixes with raindrops and turns them red. So the sky's strangest rain usually has an earthly source after all tiny living things or dust, carried on the wind.

Word Treasure

waterspout -- A waterspout is a spinning column of air over water that can lift things up.

updraft -- An updraft is a strong upward rush of air.

monsoon -- A monsoon is a season of heavy rain.

spores -- Spores are tiny cells that plants and algae use to spread.

alga -- An alga is a simple plant-like living thing that grows in water or damp places.
Trentepohlia -- Trentepohlia is a reddish-orange alga that grows on trees and lamp posts.

Background you need

Rain is water that falls from clouds, but strong winds can pick up tiny things -- dust, spores, or even small animals -- and carry them into the clouds.

A waterspout is a spinning column of air over water, like a small tornado, and its powerful updrafts can lift water and whatever floats in it.

Trentepohlia is a reddish-orange alga that grows on trees, rocks, and lamp posts in warm, damp places, and it releases huge numbers of tiny spores.

When wind sweeps those spores or red desert dust into rain clouds, the rain that falls can look red, which is why people long ago thought the sky itself was bleeding.

Break it down

LEAD: For thousands of years, people have reported rain that falls red -- 'blood rain.'

KEY EVENT: In 1841, strange matter fell on a tobacco field near Lebanon, Tennessee, and in 2001 crimson rain fell in Kerala, India.

+- WHO: Scientist Gerard Troost in 1841, and physicist Godfrey Louis in 2001
+- WHAT: Red rain, and the search for what causes it
+- WHEN: The Tennessee shower in August 1841; the Kerala rain during the 2001 monsoon

EVIDENCE: A government study found the Kerala red cells were spores of Trentepohlia, a reddish-orange alga.

OPEN QUESTION: Which strong wind -- a waterspout, a whirlwind, or hurricane-force winds -- dropped the 1841 shower?

Why it matters

Strange sights in the sky can be scary, but science can usually find a natural cause, which helps us understand the world instead of fearing it.

Test what you remember

1. What is 'blood rain'?

(A) Rain that falls only at night
(B) Rain that burns the skin
(C) Rain made of real blood
(D) Rain that falls red

2. What did scientist Gerard Troost find in the 1841 samples?

(A) Reddish-brown muscle fibers that looked like dried beef
(B) Spores of a reddish-orange alga
(C) Red dust from the Sahara
(D) Tiny glass beads

3. What is a waterspout?

(A) A pipe that carries rainwater
(B) A spinning column of air over water that can lift things up
(C) A storm that only happens in deserts
(D) A kind of rain cloud

4. What did Godfrey Louis say the red cells in the Kerala rain were?

(A) Drops of paint from airplanes
(B) Alien microbes released by a comet
(C) Spores of a reddish-orange alga
(D) Red dust from the Sahara

5. What did the government investigation find the red cells actually were?

(A) Pollen from pine trees
(B) Alien microbes from a comet
(C) Spores of Trentepohlia, a reddish-orange alga
(D) Blood from a wounded animal

6. What else has been found to turn rain red?

(A) Red food coloring from factories
(B) Windblown red dust, often from the Sahara
(C) Salt water from the ocean
(D) Smoke from wildfires

Answer key (try the quiz first!): 1: D 2: A 3: B 4: B 5: C 6: B

Questions to think about

1. Godfrey Louis, physicist: Argued the red cells in the Kerala rain were alien microbes from a comet.

2. The government-commissioned investigation: Found the red cells were spores of Trentepohlia, a reddish-orange alga from nearby.

3. Gerard Troost, scientist in 1841: Guessed a powerful wind had swept up part of a decomposing animal into a storm cloud.

MY THOUGHTS (at least 20 words):